



RAMCO INSTITUTE OF TECHNOLOGY

Approved by AICTE, New Delhi & Affiliated to Anna University

Accredited by NAAC & An ISO 9001:2015 Certified Institution

NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Electronics and Communication Engineering

Academic Year 2023 – 2024 (Odd Semester)

Degree, Semester & Branch: B.E, VII & ECE A

Course Code & Title: EC8751 & Optical Communication

Name of the Faculty member (s): Dr.A.Lakshmi

Active Learning Description

- **Unit / Topic:** Unit-1 / Modes in planar wave guide
- **Course Outcome:** CO1
- **Topic Learning Outcome:** TLO 2
- **Activity Chosen:** Think Pair Share



A. Lakshmi
3/8/23
Signature of Faculty Member


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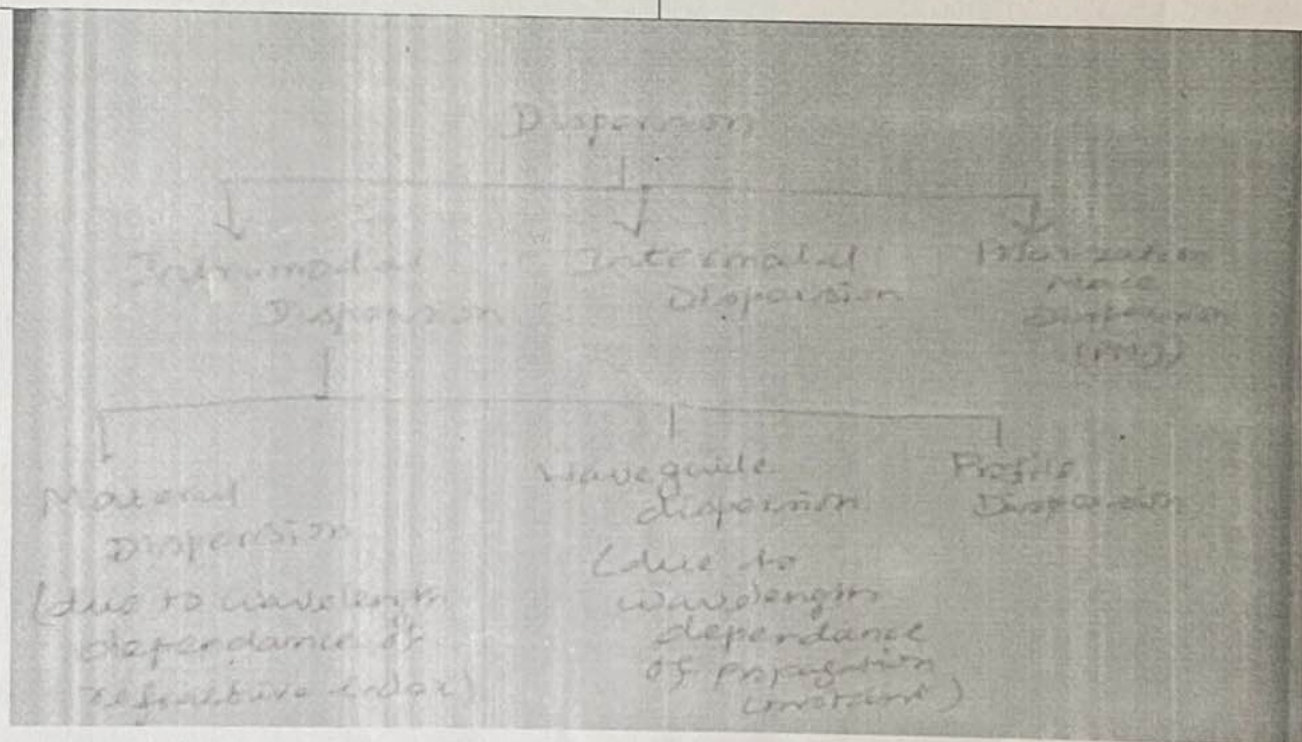
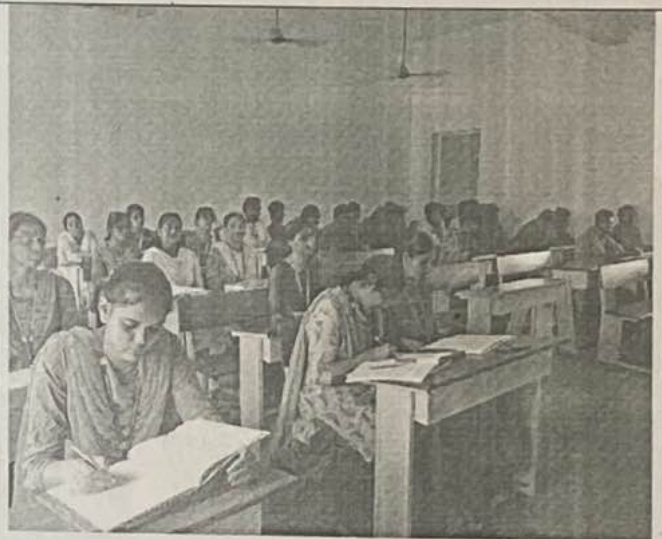
Degree, Semester & Branch: B.E, VII & ECE A

Course Code & Title: EC8751 & Optical Communication

Name of the Faculty member (s): Dr.A.Lakshmi

Active Learning Description

- Unit / Topic: Unit-2 / Dispersion optimization of single mode fiber
- Course Outcome: CO2
- Topic Learning Outcome: TLO 5
- Activity Chosen: Mind Map



A. Lakshmi
Signature of Faculty Member 10/9/23

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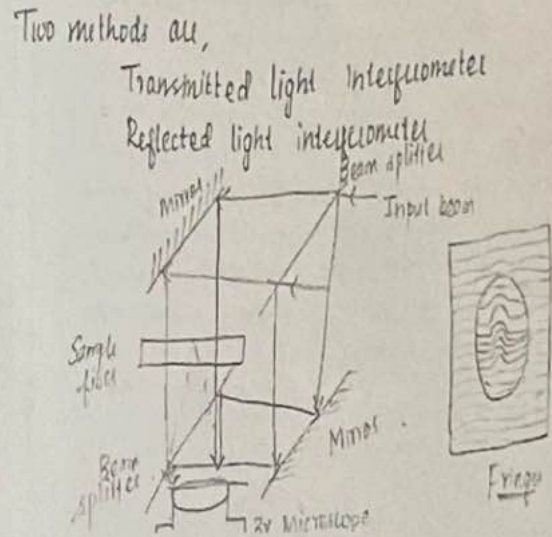
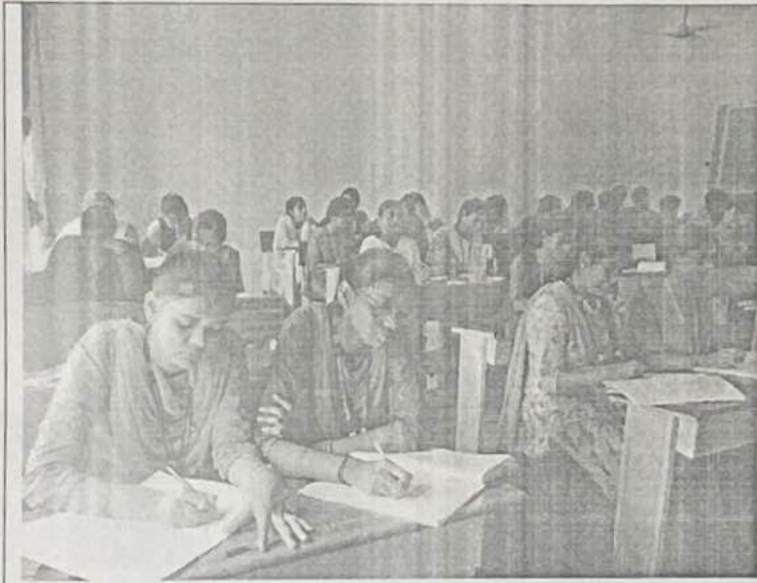
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Course Code & Title: EC8751 & Optical Communication

Name of the Faculty member (s): Dr.A.Lakshmi

Active Learning Description

- **Unit / Topic:** Unit-4 / Optical power measurement: Attenuation measurement, Dispersion measurement
- **Course Outcome:** CO4
- **Topic Learning Outcome:** TLO 11
- **Activity Chosen:** One Minute Paper



Dr. Lakshmi
3/10/23

Signature of Faculty Member

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Course Code & Title: EC8751 & Optical Communication

Name of the Faculty member (s): Dr.A.Lakshmi

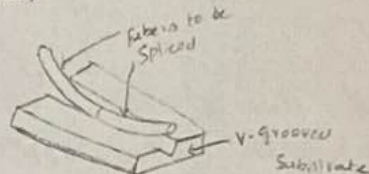
Active Learning Description

- **Unit / Topic:** Unit-4 / LED Coupling to Single Mode Fibers, Fiber Splicing, Optical Fiber connectors, Revision
- **Course Outcome:** CO4
- **Topic Learning Outcome:** TLO 12
- **Activity Chosen:** Open Book Test (Library)



V-Groove

In the V-groove splice technique, the prepared fiber ends are first buried together in a V-shaped groove. They are then bonded together with an adhesive (or) are held in place by means of cover plate. The V-shaped channel can be either a grooved silicon, plastic, ceramic (or) metal substrate. The splice loss in this method depends strongly on the fiber size and eccentricity (the position of core relative to centre of fiber).



V-groove Optical Fiber Splicing technique

A. Lakshmi
16/10/23

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Department of Electronics and Communication Engineering
Academic Year 2023 – 2024 (Odd Semester)

Degree, Semester & Branch: VII Semester B.E.ECE A
Course Code & Title: EC8751 Optical Communication
Name of the Faculty member (s): Dr.A.Lakshmi

Innovative Practice Description

- **Unit / Topic:** Unit V / SONET/SDH Rings and Networks
- **Course Outcome:** CO 5
- **Topic Learning Outcome:** TLO 14
- **Activity Chosen:** Reciprocal Peer Questioning
- **Justification:**

The students will be able to understand the concepts of SONET/SDH Rings and Networks with the help of Reciprocal Peer Questioning activity more easily. The students will be able to easily remember the concepts if questions are being asked to them.

- **Time Allotted for the Activity:** 10 minutes
- **Details of the Implementation:**

Reciprocal Peer Questioning is an activity in which students question each other about the content they are learning using higher-order, open-ended question stems. The questions are used to promote thinking and generate focused discussions in small groups.

The students were in the class room. First the students were taught with the concepts of SONET/SDH Rings and Networks. Then the students were allowed to discuss the topics among their friends. Then the students were encouraged to ask questions to each other by which the concepts are learnt easily and more quickly.

- **CO – PO / PSO mapping:**

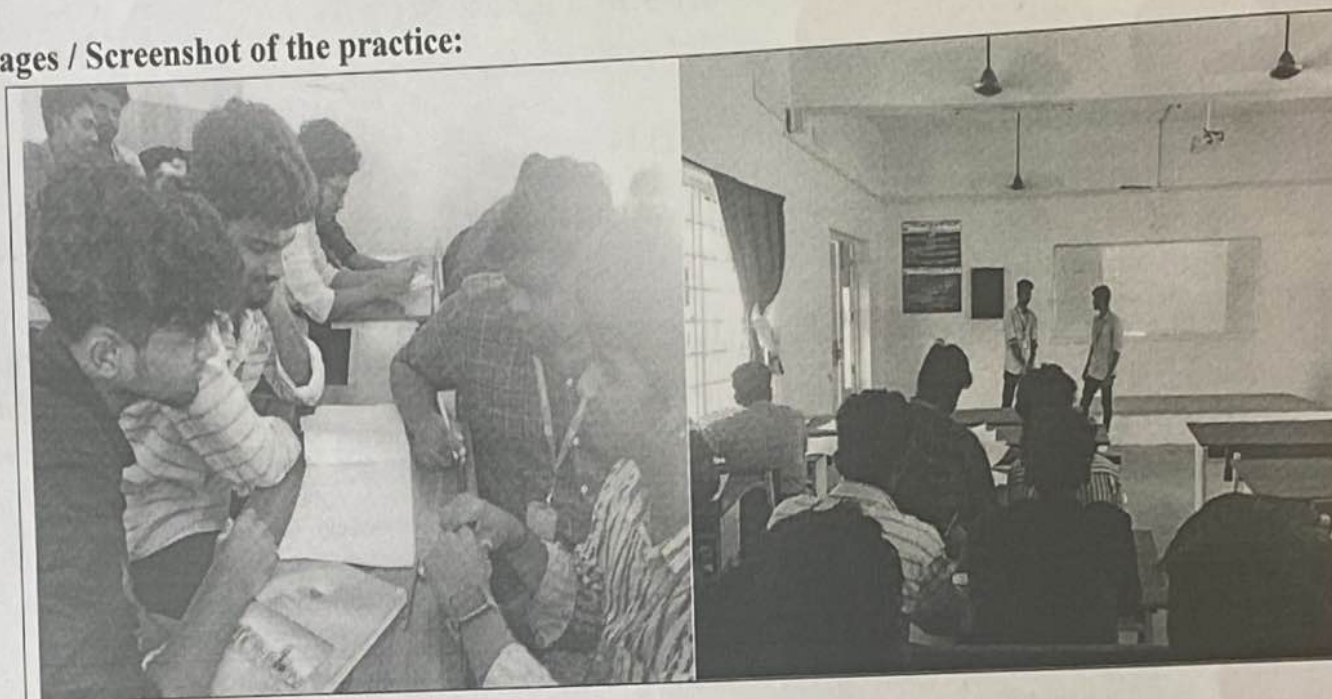
CO	PO1	PO2	PO3	PO5	PO9	PO10	PO12	PSO3
CO5	3	2	3	3	2	2	1	3

(1 – Low 2 – Moderate 3 – High)

• PO / PSO mapped:

Innovative practice	PO9	PSO3
	2	3
Justification for correlation	Reciprocal Peer Questioning activity on SONET/SDH networks helps the students to present results as a team, with smooth integration of contributions from all individual efforts.	The knowledge of SONET/SDH optical architecture, WDM, Passive WDM components is useful in designing Communication systems.

• Images / Screenshot of the practice:



• Reflective Critique:

❖ *Feedback of practice from students and other stakeholders:*

The effectiveness of the activity was felt when the students were asked the questions on the topic, they were able to answer easily. They also understood the concepts more easily and able to remember while writing the test.

❖ *Benefit of the practice:* (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

The students were more interested in learning the concepts if they are asking questions and sharing the concepts through their friends. They clarified their doubts with the neighboring students and the faculty. They learnt the concepts and asked more doubts in the topic and learnt many things. Outcome attainment has also been increased since the students' performance in the test is also good.

❖ **Challenges faced in implementation:**

Basically some time is needed to explain the basic concepts of SONET/SDH to the students and also to discuss among the students.

References:

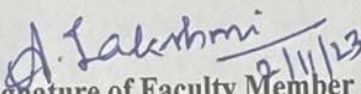
Textbooks:

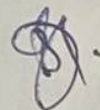
1. P Chakrabarti, "Optical Fiber Communication, McGraw Hill Education (India) Private Limited, 2016 (UNIT I, II, III)
2. Gerd Keiser, "Optical Fiber communication" MC Graw -Hill international, 4th Edition, 2010

Other References:

Books / Website/ Journal/e-books/ Videos/ NPTEL/ Matlab Online/ Online Course etc.

1. John M. Senior, "Optical Fiber communication", Second Edition, Pearson Education, 2007
2. Ramaswami, Sivarajan and sasaki "Optical Networks ", Morgan kaufmann, 2009
3. J. Senior, " optical communication, principles and practice", Prentice Hall of India, 3rd Edition, 2008.
4. NPTEL video on "Advanced Optical Communication"
Link: <https://nptel.ac.in/courses/117/101/117101002/>


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Department of Electronics and Communication Engineering
Academic Year 2023 – 2024 (Odd Semester)

Degree, Semester & Branch: VII Semester B.E.ECE A

Course Code & Title: EC8751 Optical Communication

Name of the Faculty member (s): Dr.A.Lakshmi

Innovative Practice Description

- **Unit / Topic:** Unit III / Laser structures and radiation patterns, Single mode laser, External modulation-temperature effort
- **Course Outcome:** CO 3
- **Topic Learning Outcome:** TLO 9
- **Activity Chosen:** In Class Teams
- **Justification:**

The comparison between all the structures of LASER structure is generic and only a small difference in the structures. Through the In class Teams activity the individual student will learn the concepts and structural difference clearly as they discussed with the peer group members. Hence I decided to conduct this activity for better understanding of the concept.
- **Time Allotted for the Activity:** 20 minutes (10 minutes for Teaching and 10 minutes for response & discussion between teams.
- **Details of the Implementation:**

In Class Teams is a collaborative teaching strategy. It can be used to help students form individual ideas, discuss and share with the others in-group. It can be used before reading or teaching a concept and works better with smaller groups.

The students were in the class room. First the students were taught with the concepts of Laser structures and radiation patterns, Single mode laser, External modulation-temperature effort. Then the students were allowed to discuss the topics among their friends. Then two students grouped and shared their views on the topic. Since the students discussed among their friends, they understood the concepts more clearly.

• **CO – PO / PSO mapping:**

CO	PO1	PO3	PO9	PO10	PSO1	PSO3
CO3	3	3	2	2	3	3

(1 – Low

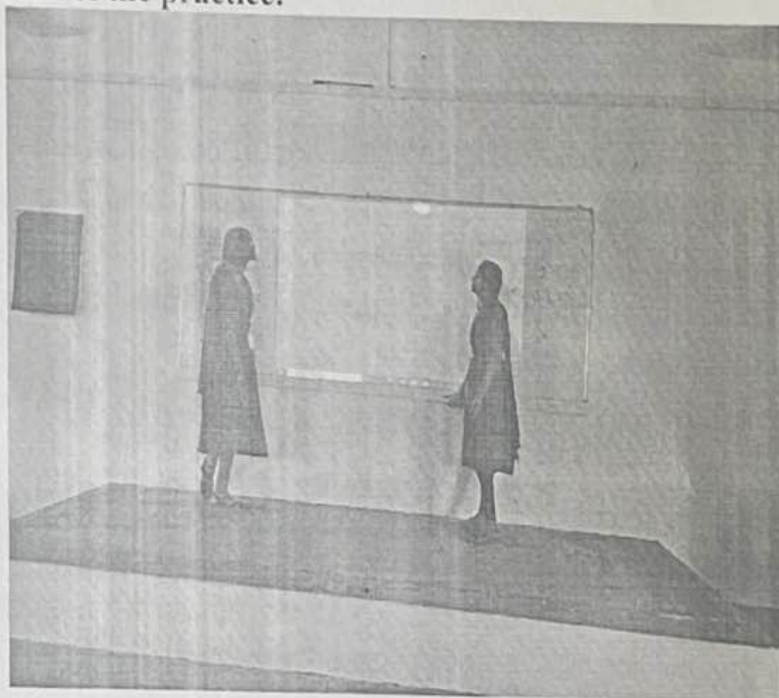
2 – Moderate

3 – High)

- PO / PSO mapped:

Innovative practice	PO9	PO10	PSO3
	2	2	3
Justification for correlation	In-Class Teams activity helps the students to present results as a team, with smooth integration of contributions from all individual efforts.	The topic deals with optical sources like LED, Laser to find internal and external quantum efficiency. This task is accomplished through team activity.	To analyze the various optical sources, laser structures, radiation patterns & External modulation-temperature effort, classify various types of detectors and to design & compare optical detectors, this activity is really useful.

- Images / Screenshot of the practice:



- Reflective Critique:

❖ *Feedback of practice from students and other stakeholders:*

The effectiveness of the activity was felt when the students were asked the questions on the topic, they were able to answer easily. They also understood the concepts more easily and able to remember while writing the test.

❖ *Benefit of the practice:* (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

The students were more interested in learning the concepts if they are sharing the concepts through their friends. They clarified their doubts with the neighboring students and the faculty. They learnt the concepts and asked more doubts in the topic and learnt many things. Outcome attainment has also been increased since the students' performance in the test is also good.

❖ *Challenges faced in implementation:*

Basically some time is needed to explain the basic concepts of laser structures, radiation patterns to the students and also to discuss among the students.

References:

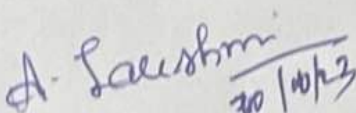
Textbooks:

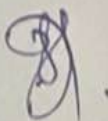
1. P Chakrabarti, "Optical Fiber Communication, McGraw Hill Education (India) Private Limited, 2016 (UNIT I, II, III)
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